UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2023/2024 FIRST SEMESTER EXAMINATION

COURSE:	MTH314: COMPLEX VARIABLES I
DATE:	MONDAY 3RD FEBRUARY, 2025
TIME ALLOWED:	TWO HOURS (2 HRS)

Instructions: Attempt any four (4) questions only.

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1. (a) When is a function, $f(z)$, said to be analytic?
   (b) State the necessary and sufficient conditions for a function, $f(z)$, to be analytic.
   (c) Prove the Cauchy-Riemann equations

2. (a) Given that $f(z) = x + iy$, define the following
   (i) Limit of the function $f(x)$
   (ii) Continuity of the function $f(x)$
   (b) Find the value(s) of z such that $e^z = -1$
   (c) Prove that $sin^2 z + cos^2 z = 1$.

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11.6

3. (a) Define harmonic function $f(z)$.
   (b) Prove that if $f(z) = u + iv$ is an analytic function, then u and v are both harmonic.
   (c) Show that the function $u(x, y) = y^3 - 3x^2y$ is harmonic and find its harmonic conjugate.

11.6

4. (a) State the Cauchy integral theorem.
   (b) Verify that the real and imaginary parts of $f(z) = z^2 + 5iz + 3 - i$ satisfy the $C - R$ equations and deduce the analyticity of the function.
   (c) Evaluate $\lim \dfrac{z^2 + 4z + 3}{z + 1}$ as $z \to (1 - i)$

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5. (a)Define (i) Jordan arc (ii) Contour (iii) Zeros of an analytic function
   (b) Examine the continuity of the function
   $f(z) = \{\dfrac{z^3 - iz^2 + z - i}{z - i}, z \neq i$ and $0, z = i$

6. (a) Define the following:
   (i) Simply Connected Region (ii) Multi-connected Region (iii) Single-valued and multi-valued functions.
   (b) Find the imaginary part of the analytic function whose real part is $x^3 - 3xy^2 + 3x^2 - 3y^2$

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